Beyond WER: Quantifying ASR Error Propagation in Spoken Language Understanding for Dysarthric Speech

Zixuan Yin (Presenter)[1], Xiuwen Zheng[1], Haolong Zheng[1], Kush Bhardwaj[1], Mark Hasegawa-Johnson[1]

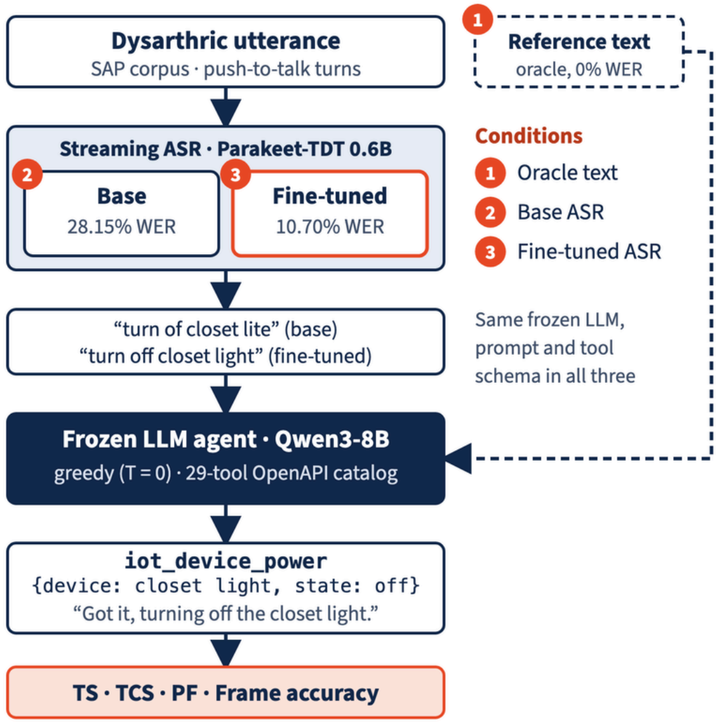
[1] University of Illinois Urbana-Champaign · Speech Accessibility Project, Statistical Speech Technology Lab

1. Introduction

- Dysarthria pushes commercial ASR WER from 3.2% to **36.3%**
- One wrong phoneme changes the meaning: “call Kathy” → “call Cassie”, “turn on” → “turn off”, “5mg” → “50mg”.
- Errors in ASR-sensitive slots may propagate to LLM tool execution

What does fine-tuning a streaming ASR on dysarthric speech buy us in a downstream assistant?

2. Pipeline architecture



3. SAP Schema: 29 tools, 10 domains

Derived from Amazon MASSIVE (60 intents, 18 scenarios, 55 slot types).

1. Filter MASSIVE intents and slots to daily assistive needs.
2. Construct 10 domains and 29 tools based on dataset
3. Iterate after human annotations and LLM label pilot

Accessibility	accessibility_dictation_toggle · accessibility_ui_scale
Calendar	alarm_manage · schedule_manage
Communication	call_make · call_manage · messages_manage · email_manage
General QA	qa_search · weather_query · device_find · order_manage
Health	health_medication_reminder · health_refill_query · health_info_query
Maps & Places	maps_route · places_search · reservations_manage
Media	media_play · media_search · media_control · media_volume
Notes & Memory	lists_manage · memory_note_manage
Smart Home	iot_device_power · iot_device_adjust · iot_sensor_query
Transport	rideshare_book · transit_lookup

4. Metrics

$$\mathrm{WER} = \frac{S+D+I}{N}$$

Evaluates accuracy of ASR model by comparing target to prediction transcript

$$\mathrm{PF} = \frac{\#\ \text{correct gold slots}}{\#\ \text{gold slots}}$$

Evaluates parameter filling (how well LLM fills slot values for tool call)

5. SAP evaluation corpus

124	34,577	1,108
speakers	recordings	benchmark clips

6. Results (N = 1,108)

Condition	WER	Tool sel.	Param fill.	Frame acc.
Oracle	0.00%	95.67%	88.41%	84.03%
Base ASR	28.15%	74.55%	46.22%	**38.09%**
Fine-tuned	10.70%	92.96%	75.73%	**66.88%**
Gain	**−17.45**	**+18.41**	**+29.51**	**+28.79**

62.67% **Adaptation Recovery Rate**
(66.88 − 38.09) / (84.03 − 38.09) of the gap recovered

7. Next steps

Conversational replies

The assistant speaks back what it understood (“Got it, turning off the closet light”).

Evaluating multi-turn tasks

Elaboration
Multi-slot commands split into sub-phrases.
Focus on cumulative dialogue tracking.

“Reschedule my meeting with Fred Placid (SLICE) at 11:30, September 18th”

Repair
High-severity speakers or low ASR confidence.
Focus on proactive error correction.

“Remind me to take a break in 30 minutes”